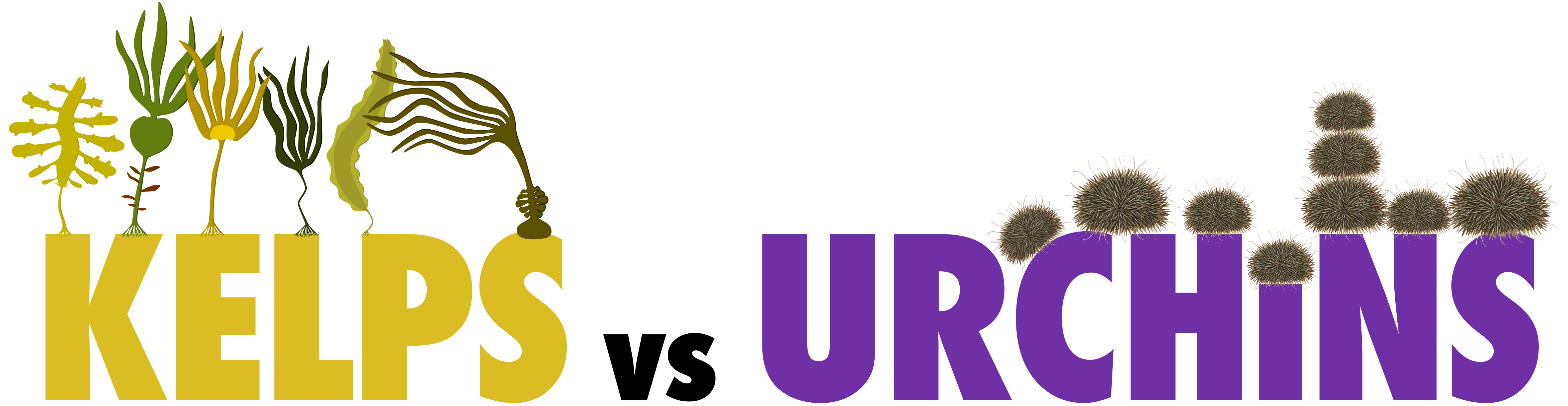




The role of sea urchins in the kelp carbon cycle

Luka Seamus Wright^{*1,2,3}, Markus Molis⁴, Juan Vicente de Miguel⁴, Anea Rakvåg⁴,
Thomas Wernberg^{1,2,5}, Karen Filbee-Dexter^{1,2,5}

¹Oceans Institute, University of Western Australia, Perth, Australia

²School of Biological Sciences, University of Western Australia, Perth, Australia

³CSIRO Oceans and Atmosphere, Perth, Australia

⁴The Arctic University of Norway, Tromsø, Norway

⁵Institute of Marine Research, His, Norway

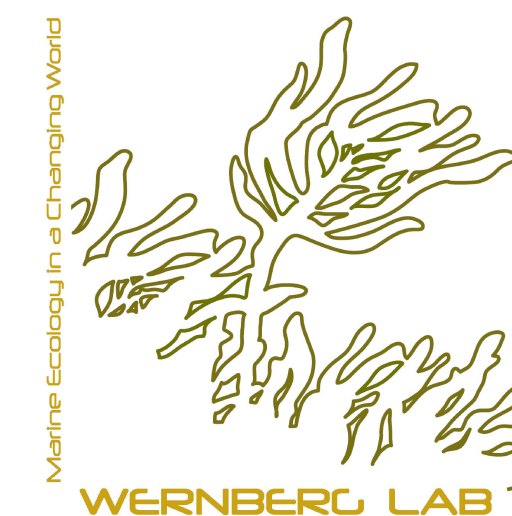
^{*}luka.wright@research.uwa.edu.au

Acknowledgements



THE UNIVERSITY OF
**WESTERN
AUSTRALIA**

**OCEANS
INSTITUTE**



**UiT The Arctic
University of Norway**



FORREST
RESEARCH
FOUNDATION

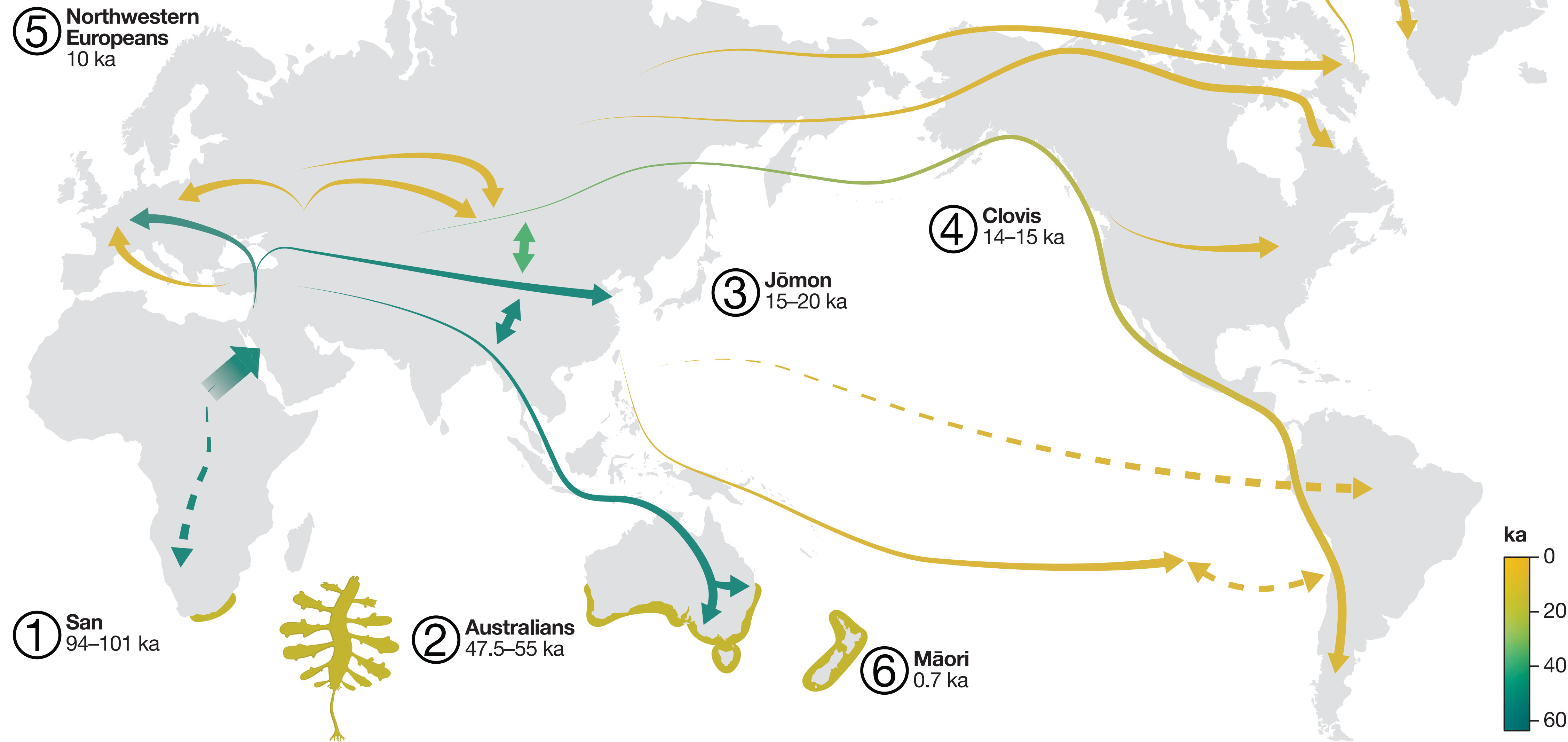


ECOLOGICAL
SOCIETY
OF AUSTRALIA



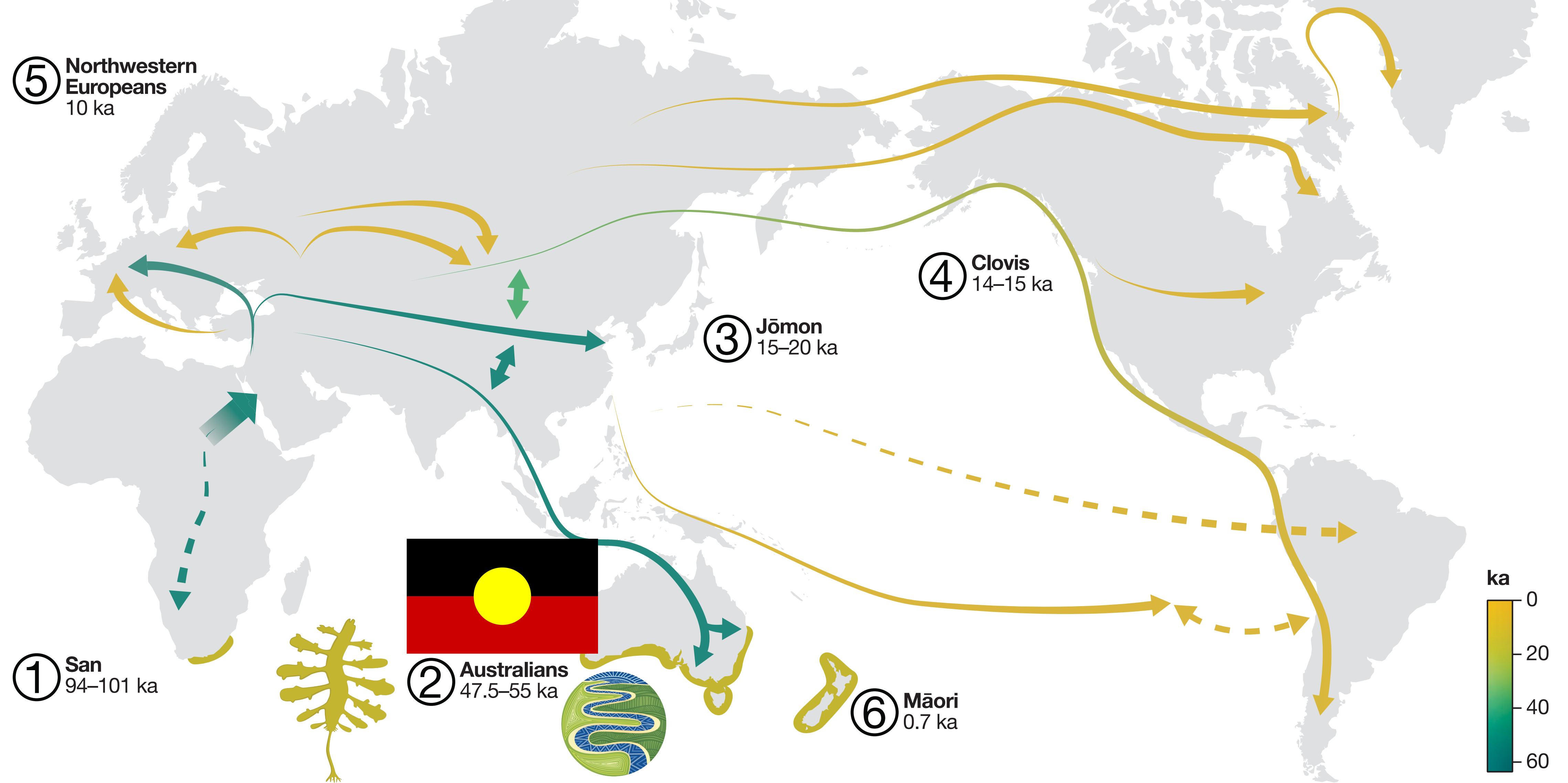
**The Research
Council of Norway**

Humans and temperate reefs



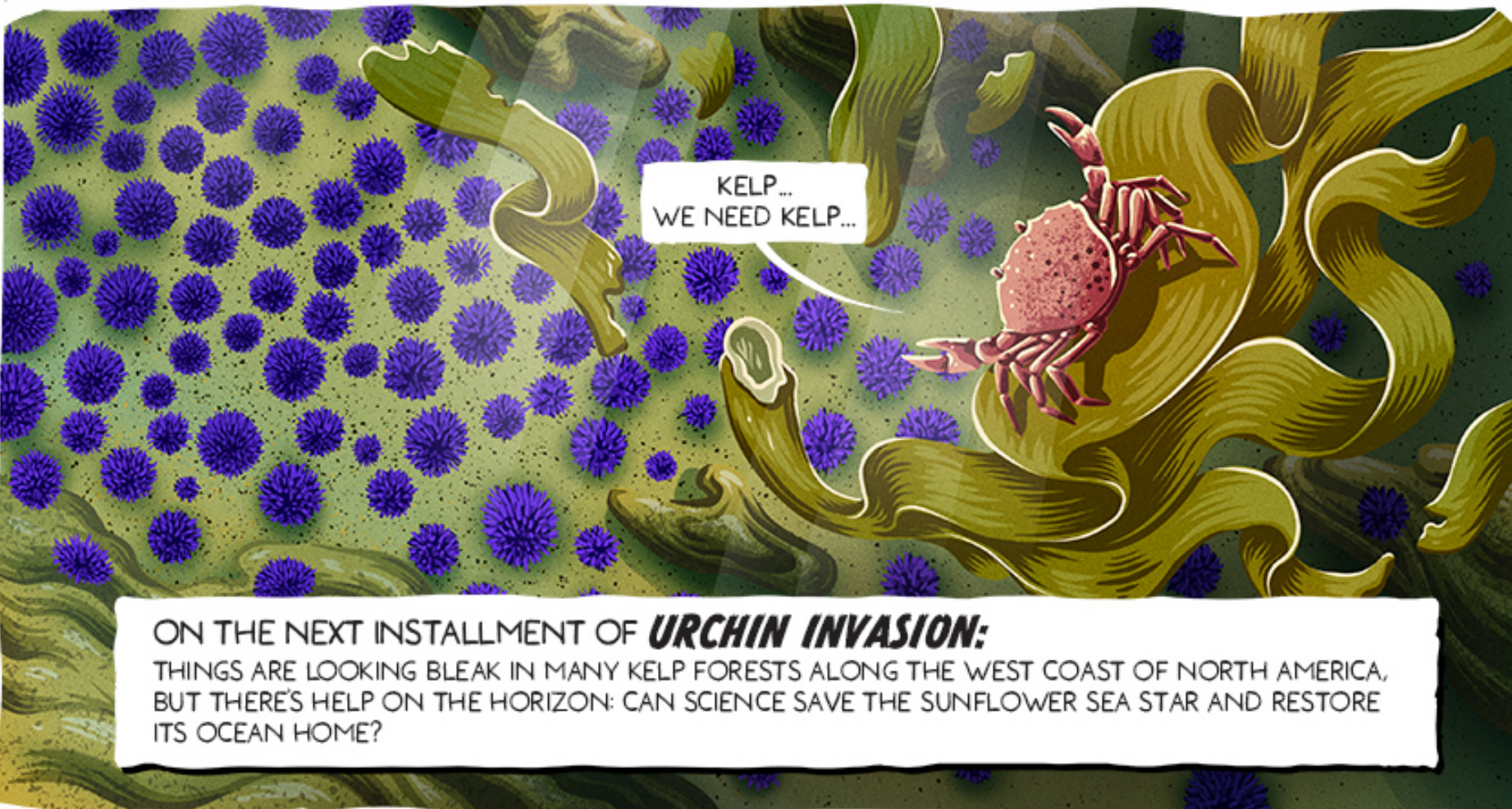
Adapted from Nielsen et al. 2017. *Nature*. doi 10.1038/nature21347.

Humans and temperate reefs

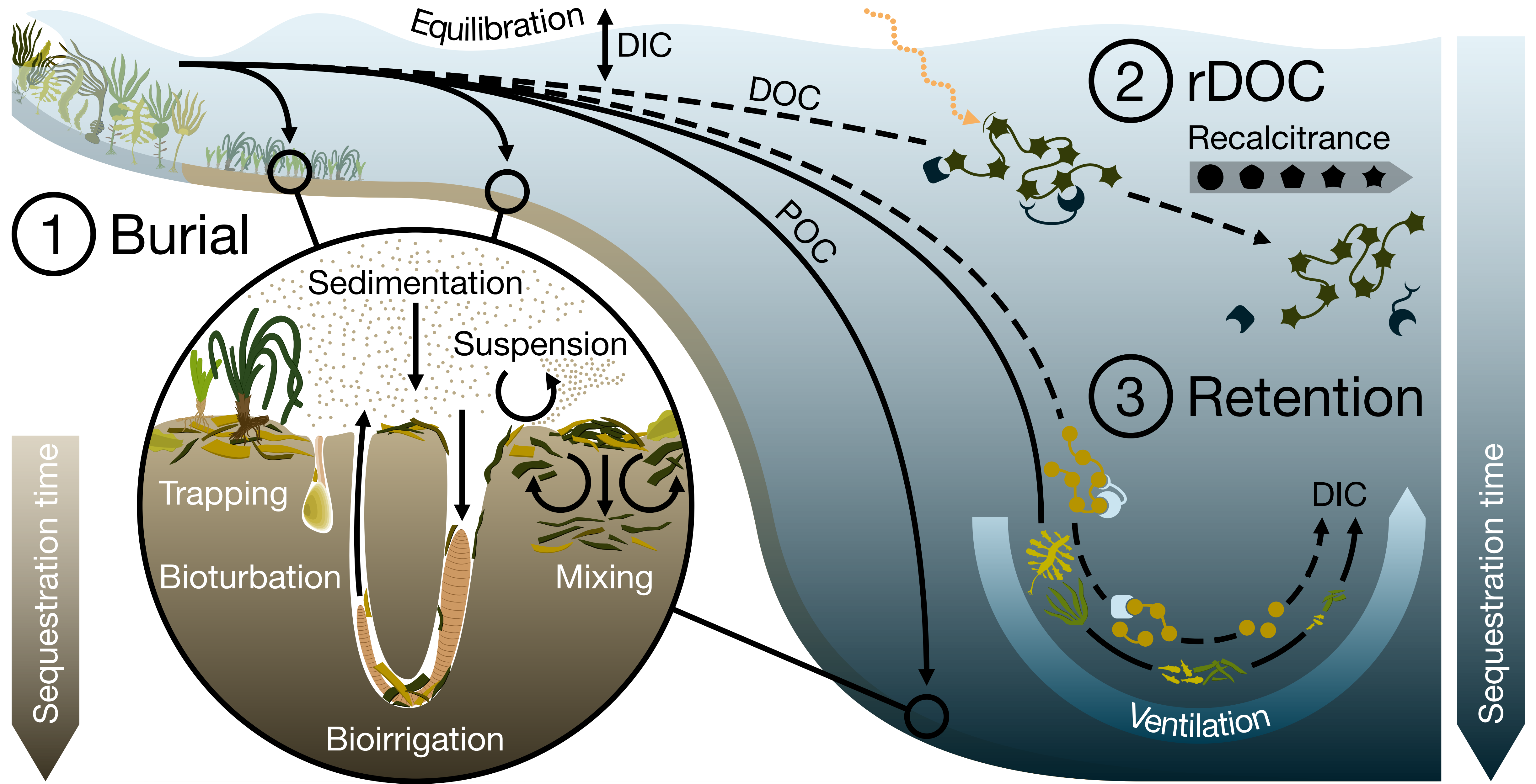


Adapted from Nielsen et al. 2017. *Nature*. doi 10.1038/nature21347.

Background



What is the role of sea urchins the kelp carbon cycle?



What is the role of sea urchins the kelp carbon cycle?
Well, duh...



But...



But...



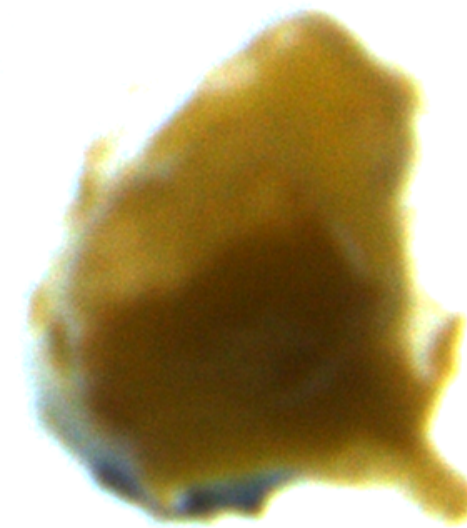
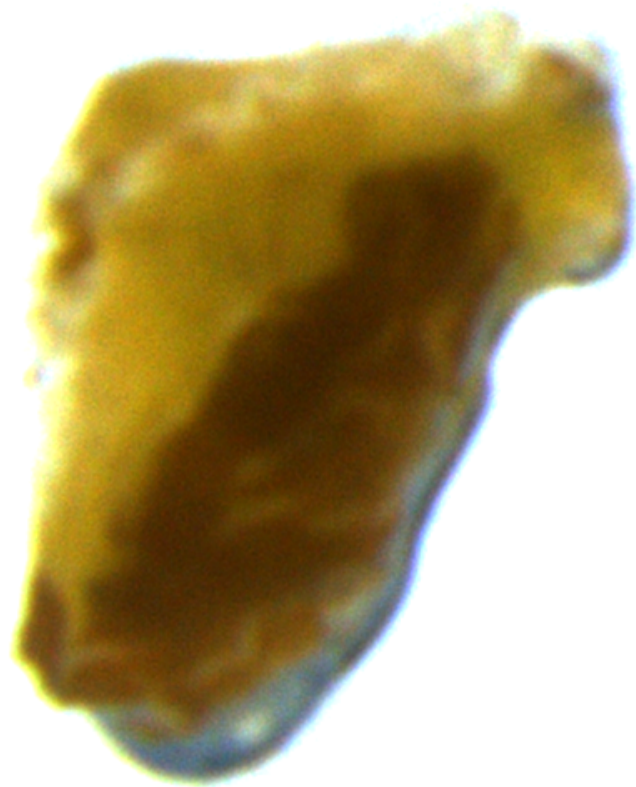
But...

Kelp confetti!



But...

Kelp confetti!



1 mm

But...

Kelp confetti!

Grazers extend blue carbon transfer by slowing sinking speeds of kelp detritus

Thomas Wernberg^{1,2}  & Karen Filbee-Dexter^{3,4}

ECOSYSTEM ECOLOGY – ORIGINAL RESEARCH



Carbon export is facilitated by sea urchins transforming kelp detritus

Karen Filbee-Dexter^{1,2}  · Morten Foldager Pedersen³ · Stein Fredriksen^{1,4} · Kjell Magnus Norderhaug¹ · Eli Rinde² · Trond Kristiansen² · Jon Albretsen¹ · Thomas Wernberg^{3,5}

1. Sinking

■ Prior ■ Kelp ■ Faeces



Data from Figure 2 in Wernberg & Filbee-Dexter 2018. *Scientific Reports*. doi 10.1038/s41598-018-34721-z.

1. Sinking

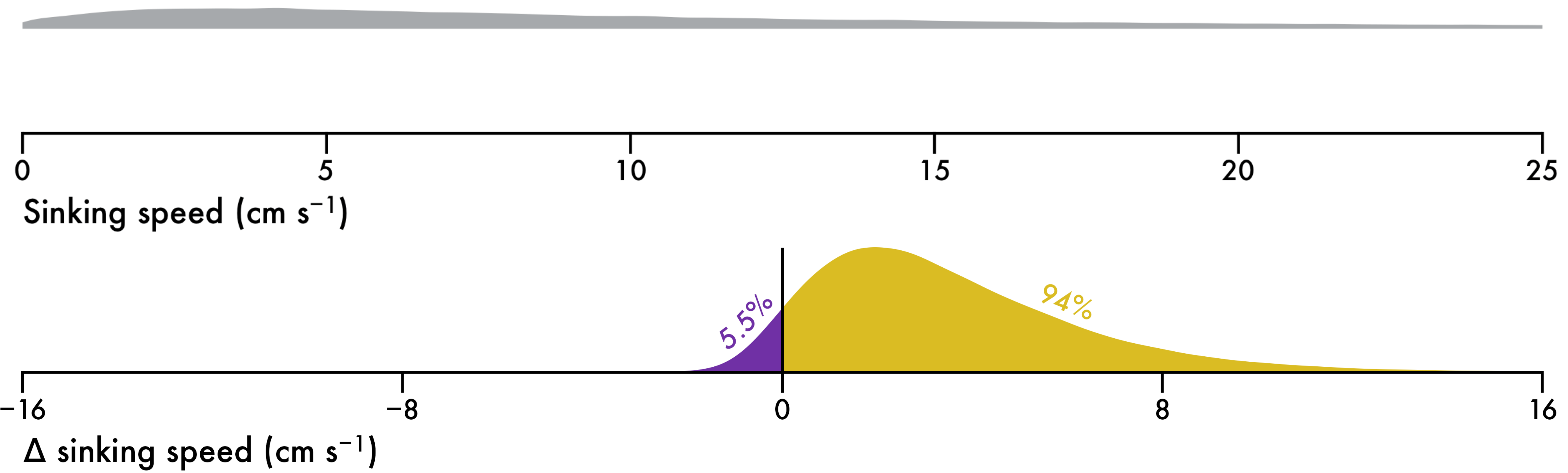
■ Prior ■ Kelp ■ Faeces



Data from Figure 2 in Wernberg & Filbee-Dexter 2018. *Scientific Reports*. doi 10.1038/s41598-018-34721-z.

1. Sinking

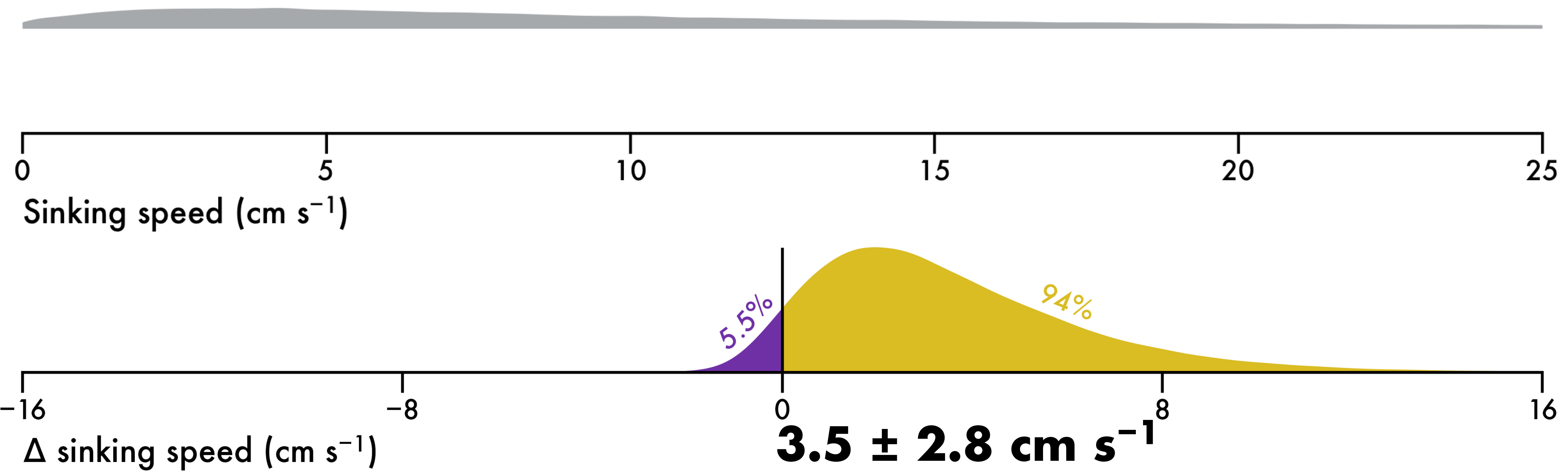
Prior Kelp Faeces



Data from Figure 2 in Wernberg & Filbee-Dexter 2018. *Scientific Reports*. doi 10.1038/s41598-018-34721-z.

1. Sinking

Prior Kelp Faeces



Data from Figure 2 in Wernberg & Filbee-Dexter 2018. *Scientific Reports*. doi 10.1038/s41598-018-34721-z.

2. Distance

■ Prior ■ Kelp ■ Faeces



Data from Figure 7 in Filbee-Dexter et al. 2020. *Oecologia*. doi 10.1007/s00442-019-04571-1.

2. Distance

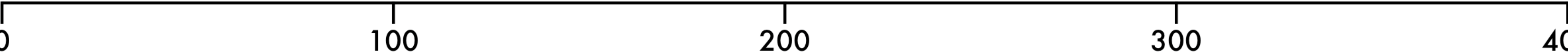
■ Prior ■ Kelp ■ Faeces



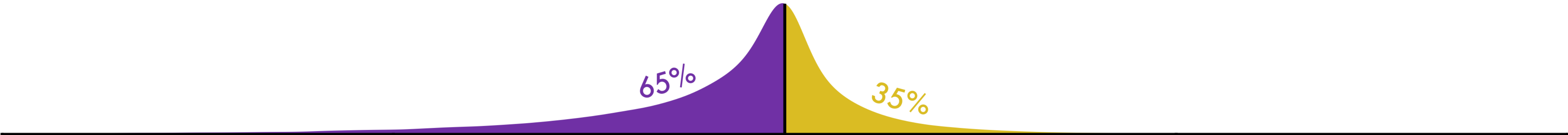
Data from Figure 7 in Filbee-Dexter et al. 2020. *Oecologia*. doi 10.1007/s00442-019-04571-1.

2. Distance

■ Prior ■ Kelp ■ Faeces



Export distance (km)

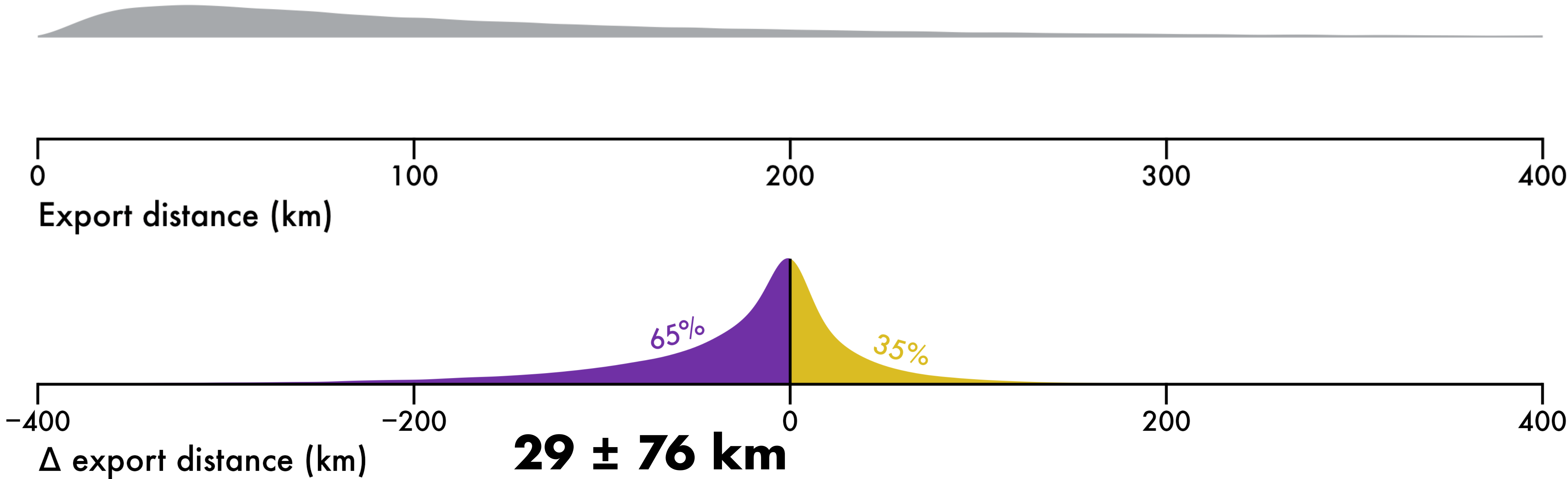


Δ export distance (km)

Data from Figure 7 in Filbee-Dexter et al. 2020. *Oecologia*. doi 10.1007/s00442-019-04571-1.

2. Distance

■ Prior ■ Kelp ■ Faeces

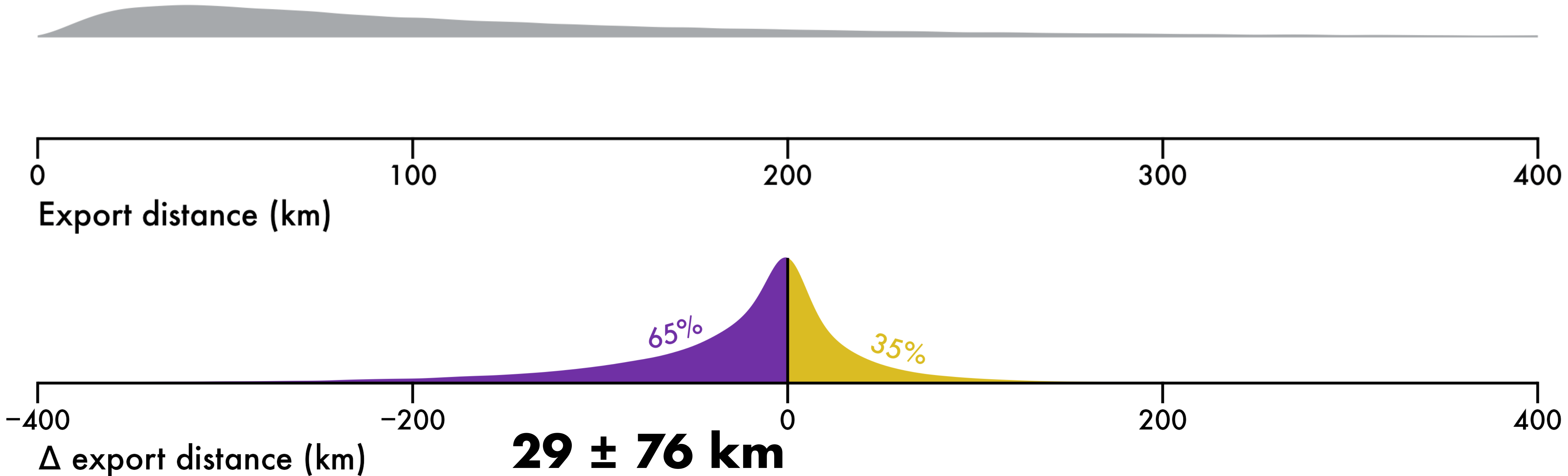


Data from Figure 7 in Filbee-Dexter et al. 2020. *Oecologia*. doi 10.1007/s00442-019-04571-1.

2. Distance

$$\text{Faeces} = 2.3 \pm 0.14 (\mu) \times \text{Kelp}$$

■ Prior ■ Kelp ■ Faeces



Data from Figure 7 in Filbee-Dexter et al. 2020. *Oecologia*. doi 10.1007/s00442-019-04571-1.

Assumptions

Assumptions

1. No carbon loss to urchin

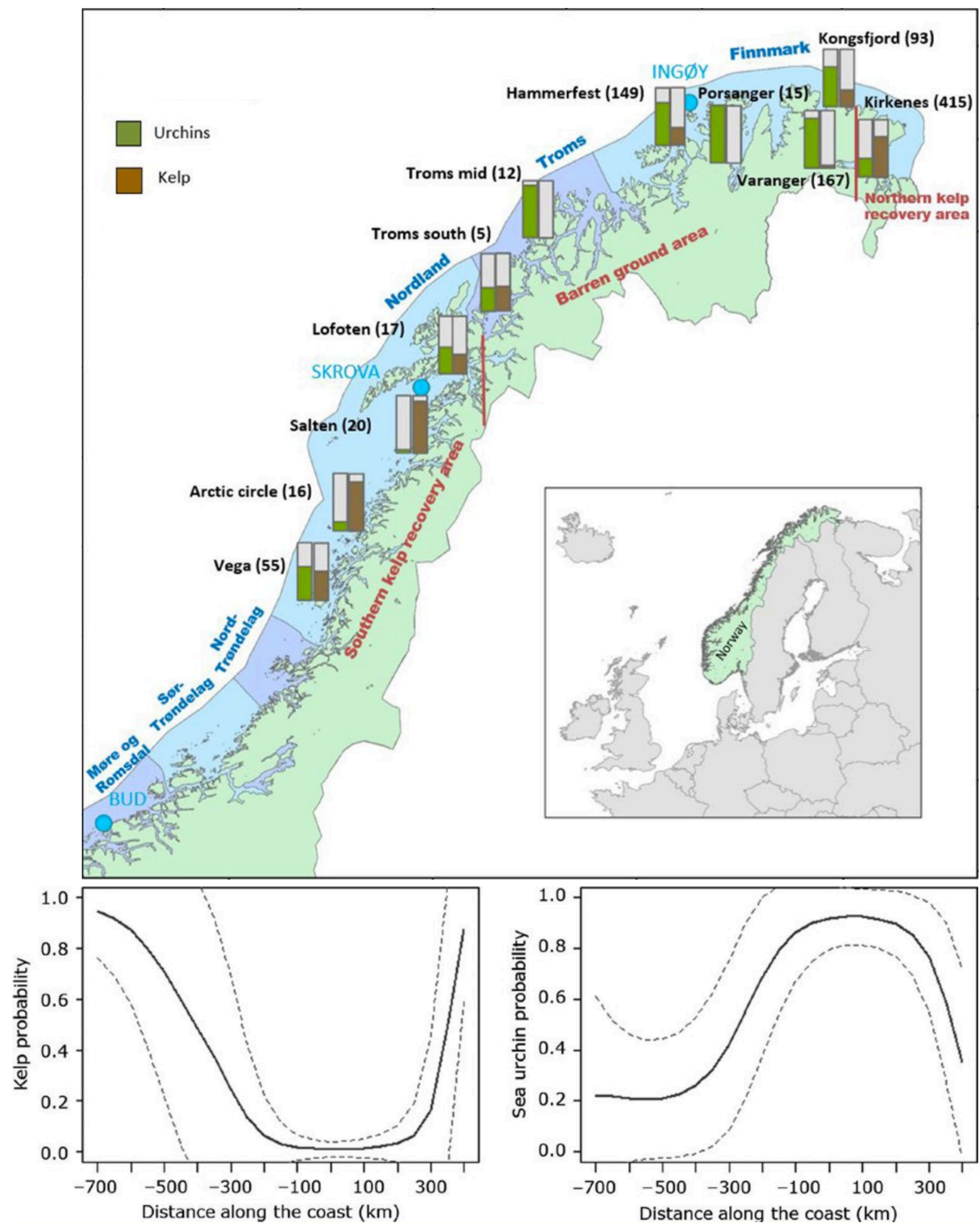
Assumptions

- 1. No carbon loss to urchin**
- 2. Identical decomposition rate**

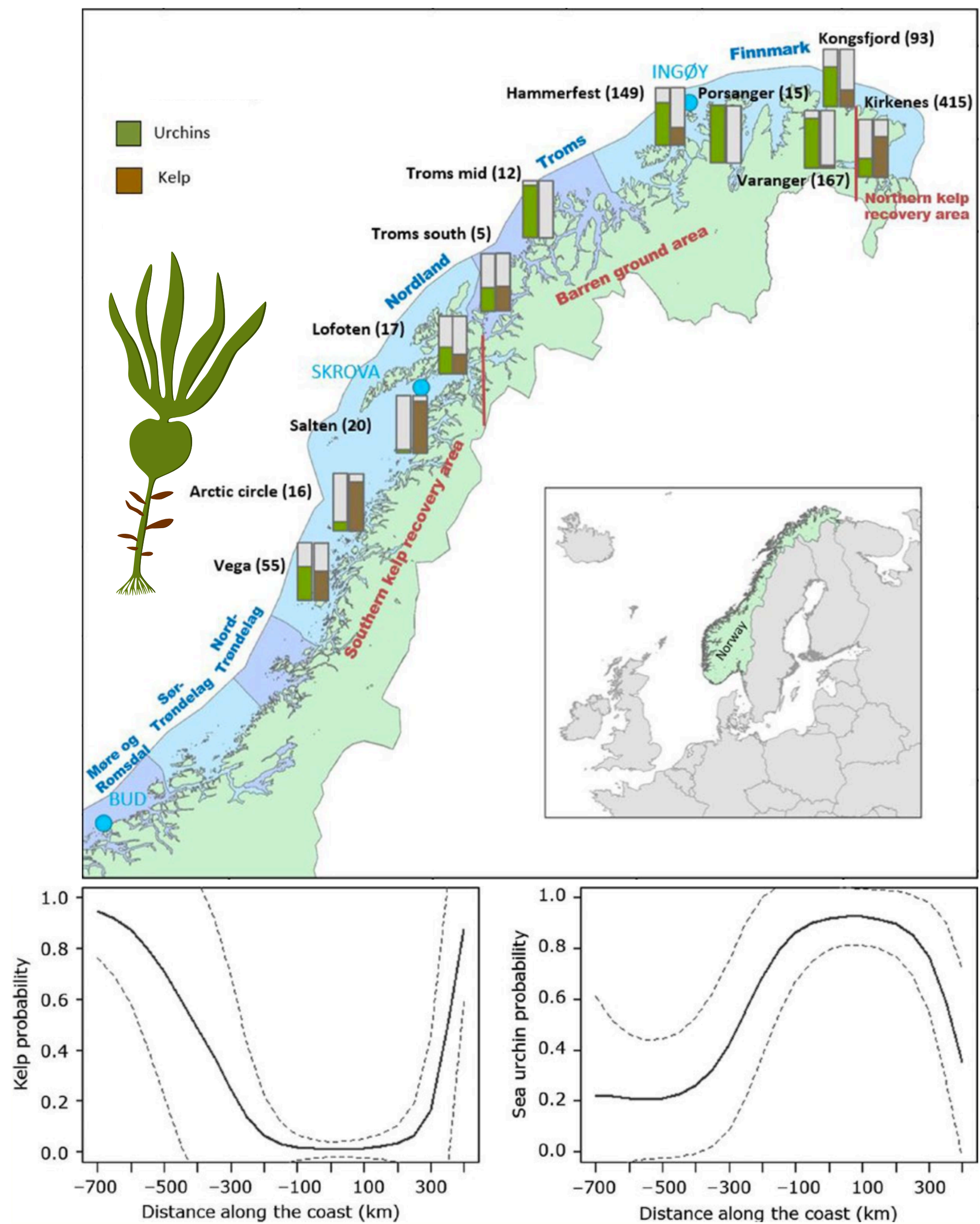
Question

Do sea urchins really extend kelp carbon transfer?

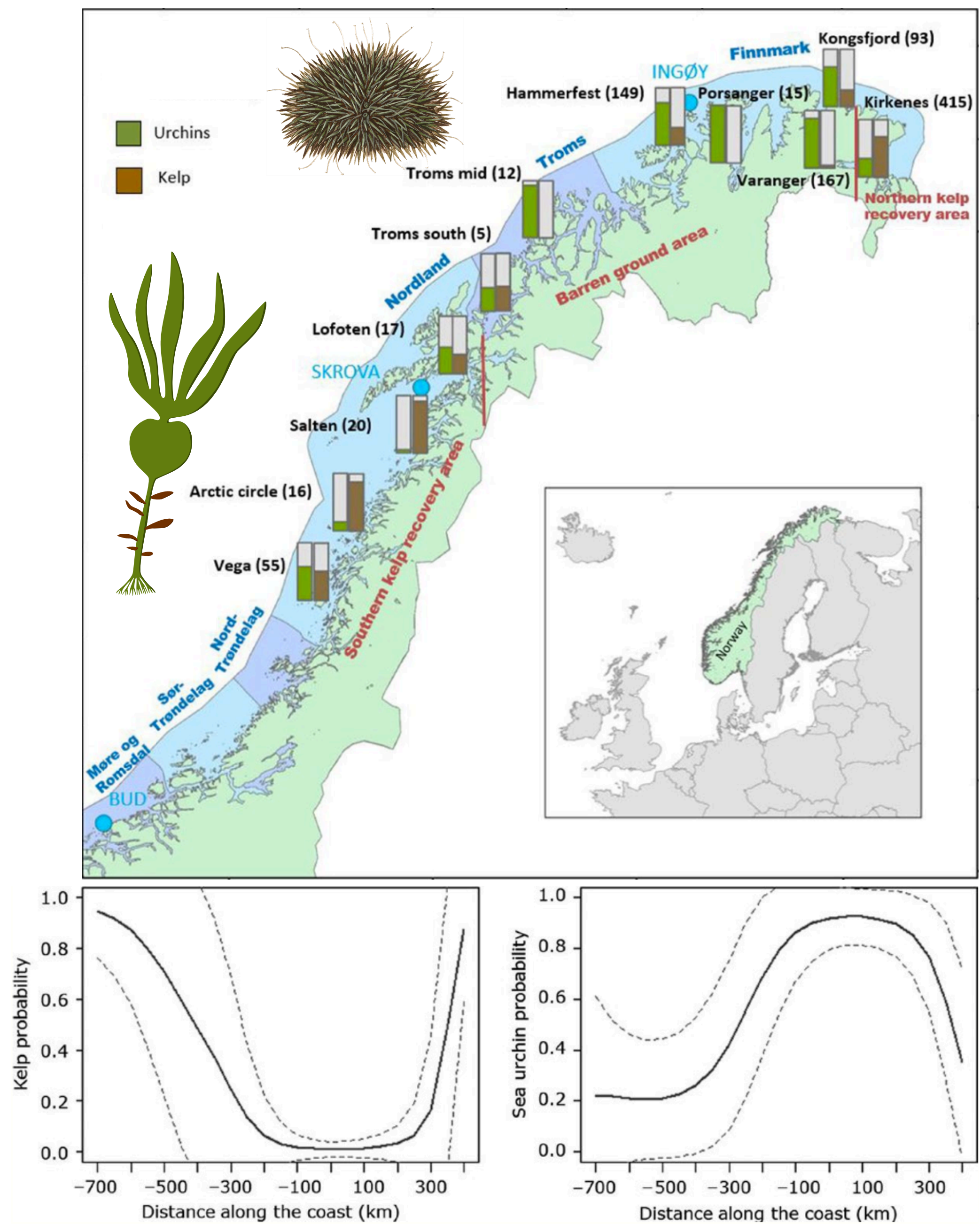
Study site



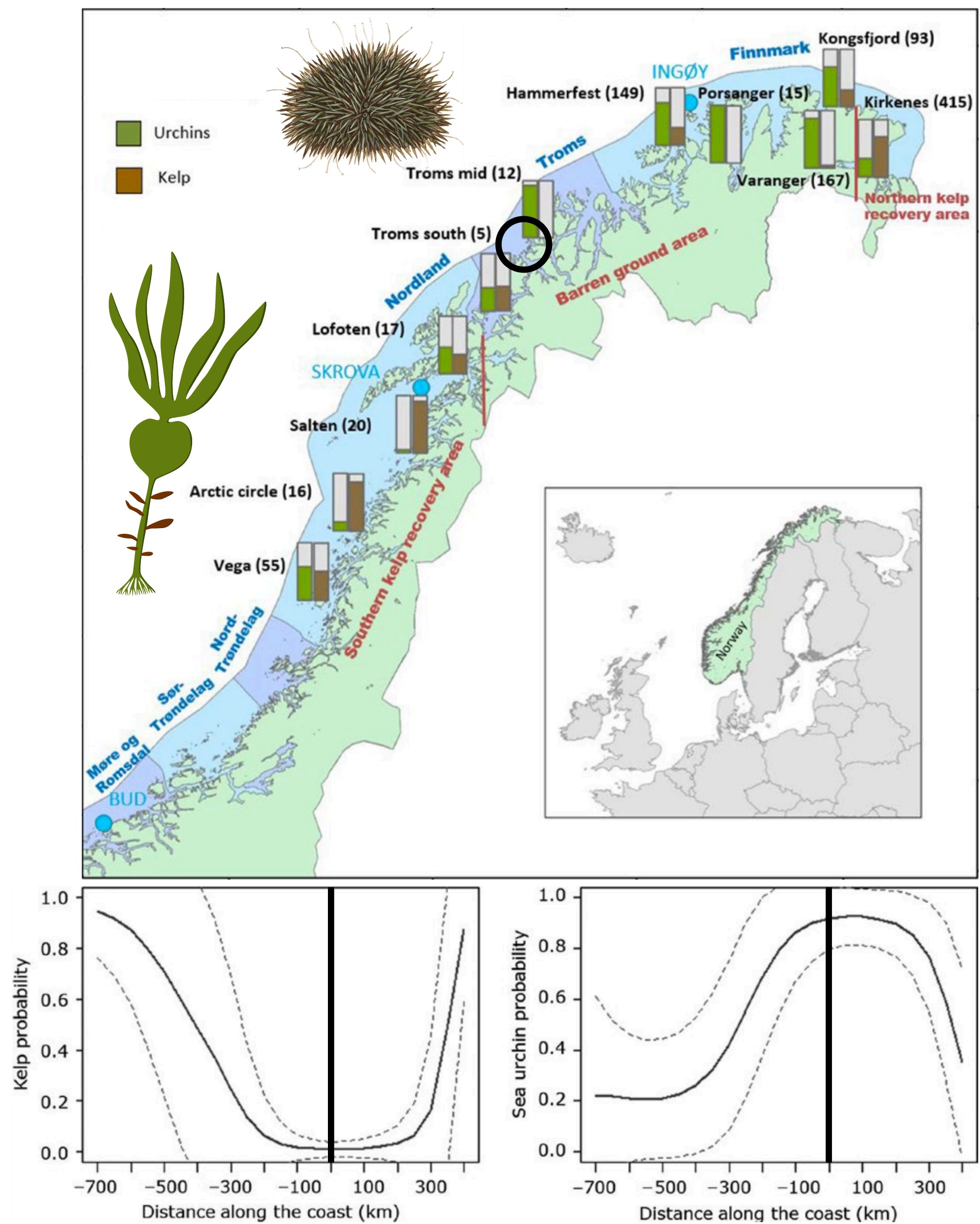
Adapted from Christie et al. 2019. *Ecology and Evolution*. doi 10.1002/ece3.4963.



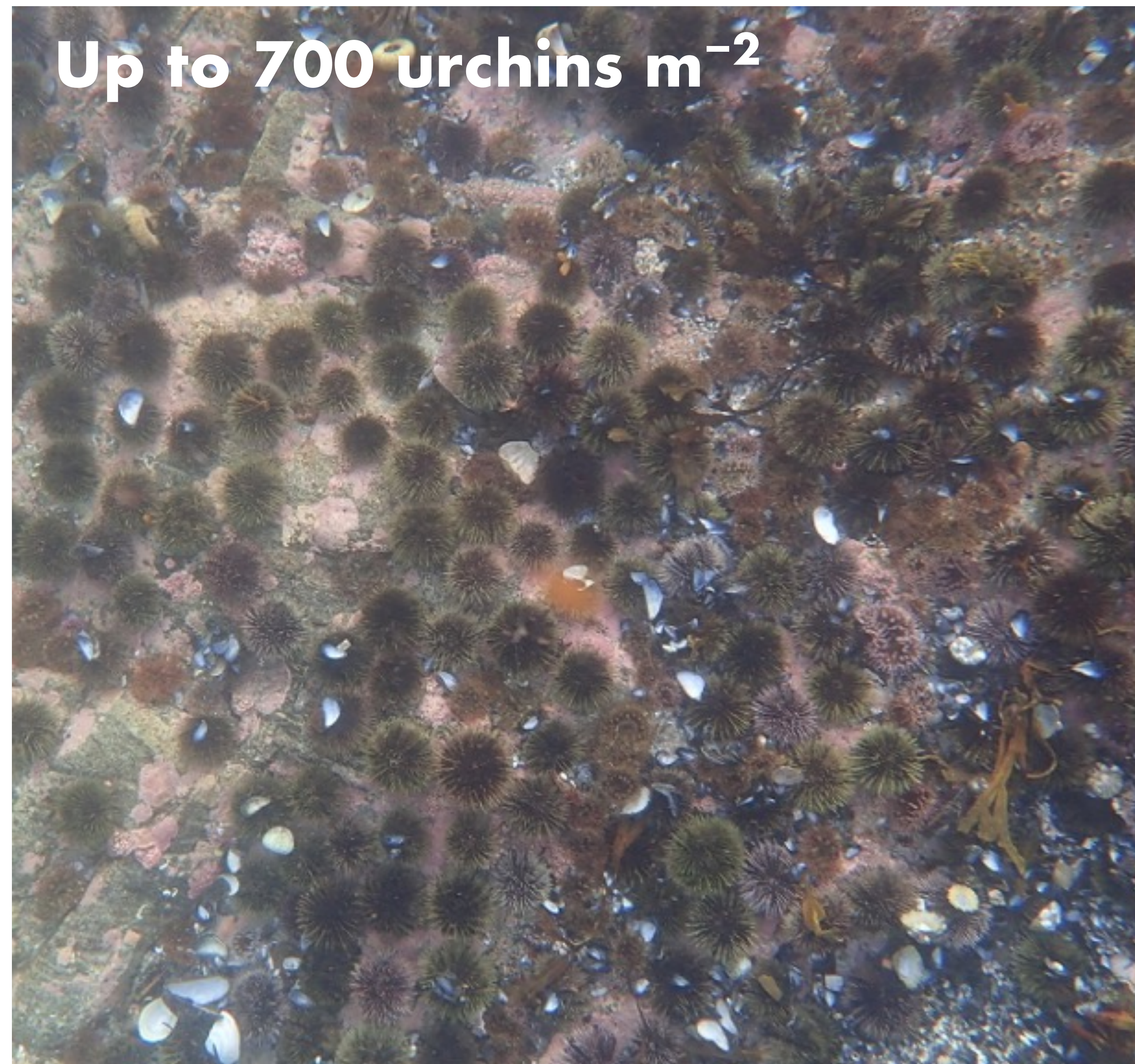
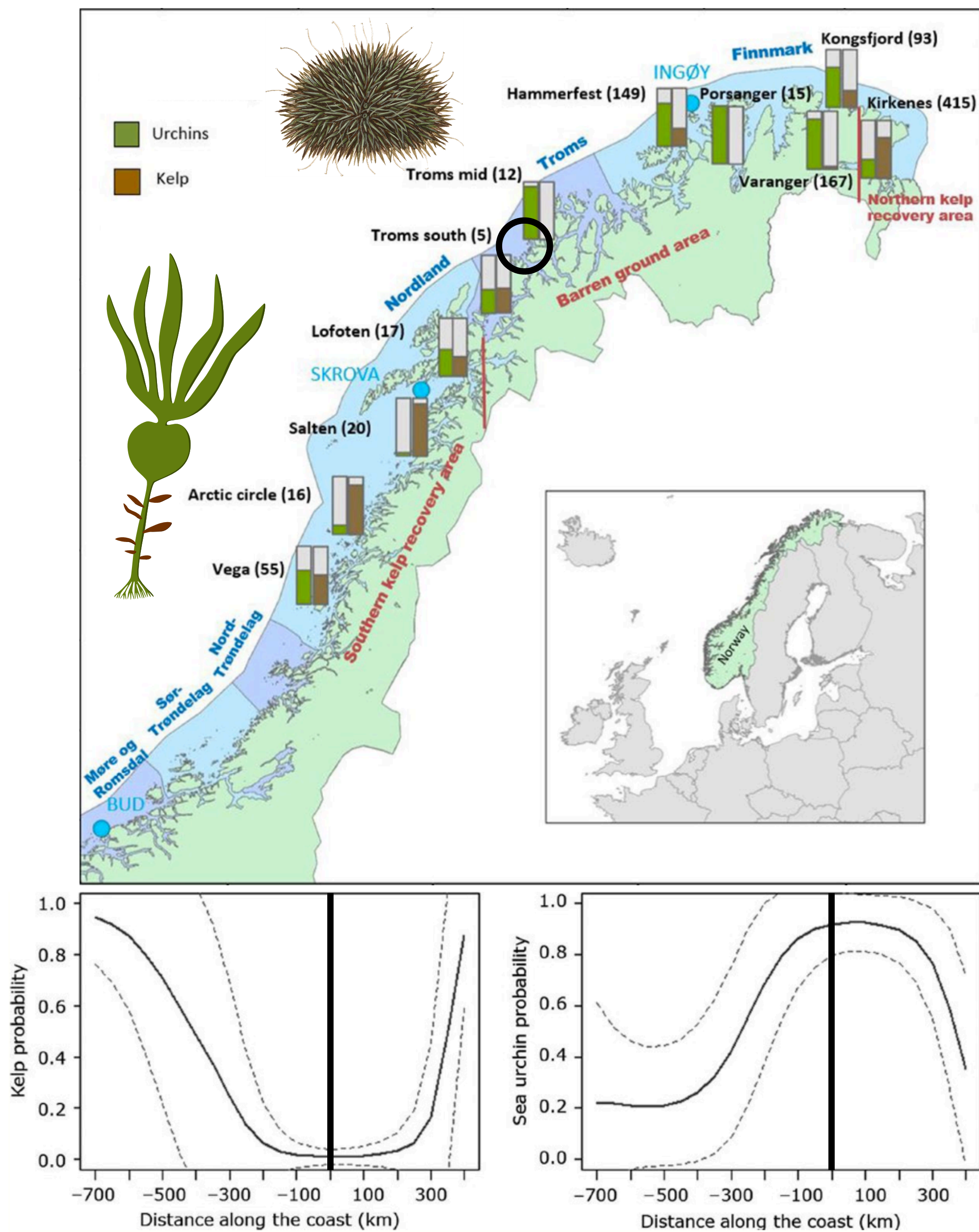
Adapted from Christie et al. 2019. *Ecology and Evolution*. doi 10.1002/ece3.4963.



Adapted from Christie et al. 2019. *Ecology and Evolution*. doi 10.1002/ece3.4963.



Adapted from Christie et al. 2019. *Ecology and Evolution*. doi 10.1002/ece3.4963.



Up to 700 urchins m⁻²

Adapted from Christie et al. 2019. *Ecology and Evolution*. doi 10.1002/ece3.4963.

Results

Results

1. Biomass

Results

1. Biomass

2. Carbon

Results

- 1. Biomass
 - 2. Carbon
- } Carbon transfer

Results

- 1. Biomass
 - 2. Carbon
 - 3. Chemical defence
- } Carbon transfer

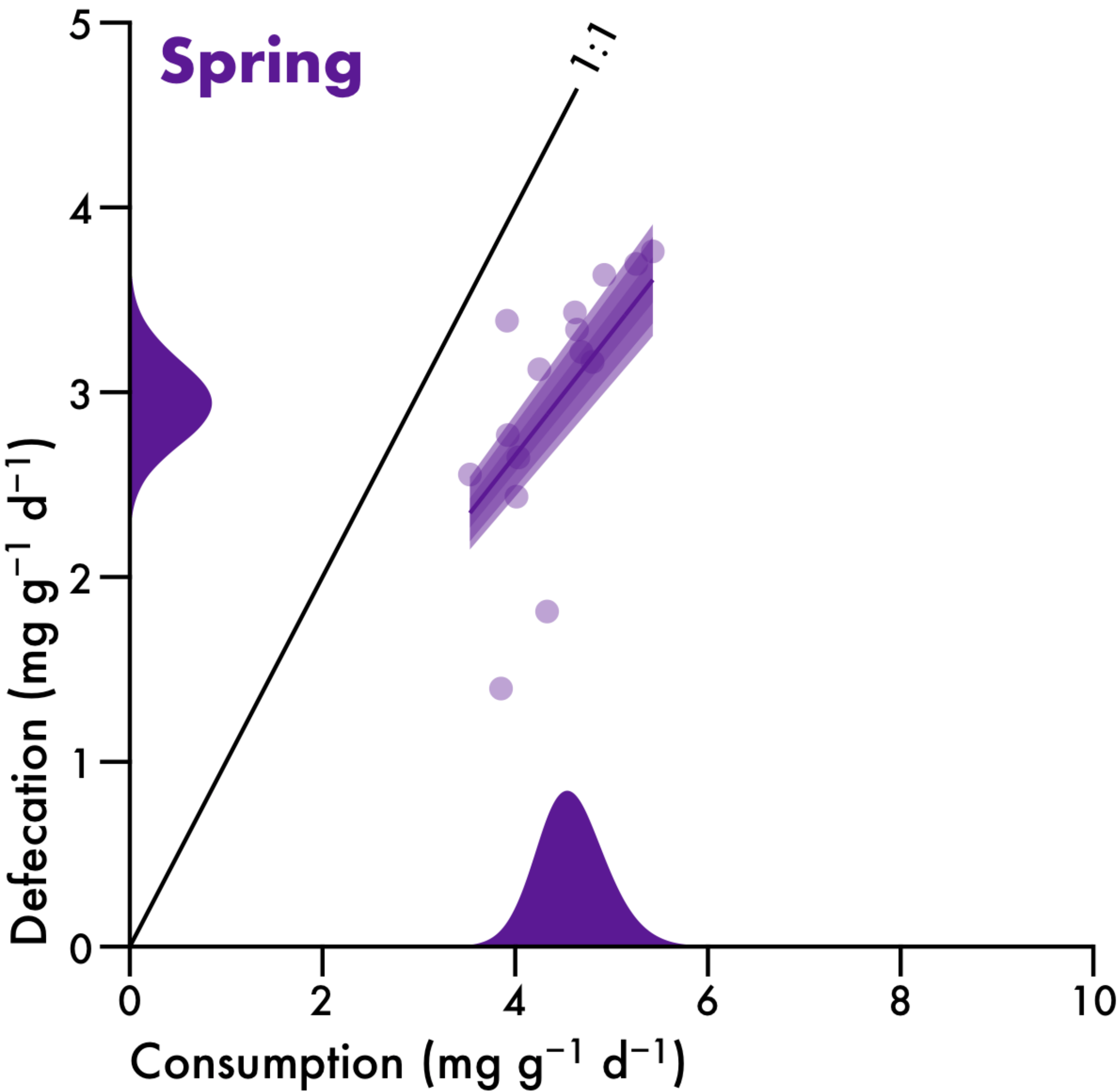
Results

- 1. Biomass**
 - 2. Carbon**
 - 3. Chemical defence**
 - 4. Photophysiology**
- } Carbon transfer**

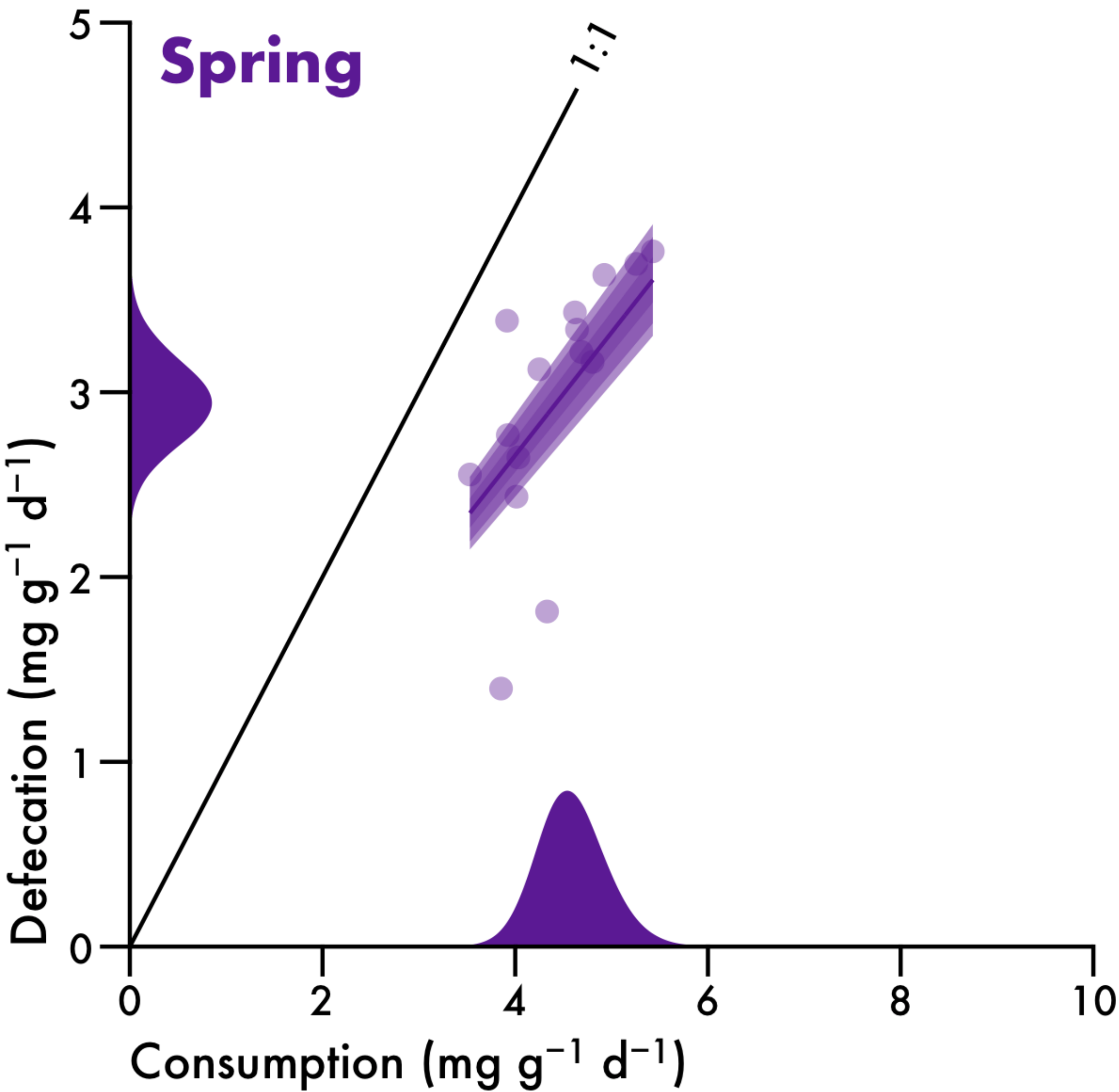
Results

- 1. Biomass
 - 2. Carbon
 - 3. Chemical defence
 - 4. Photophysiology
- } Carbon transfer
- } Decomposition

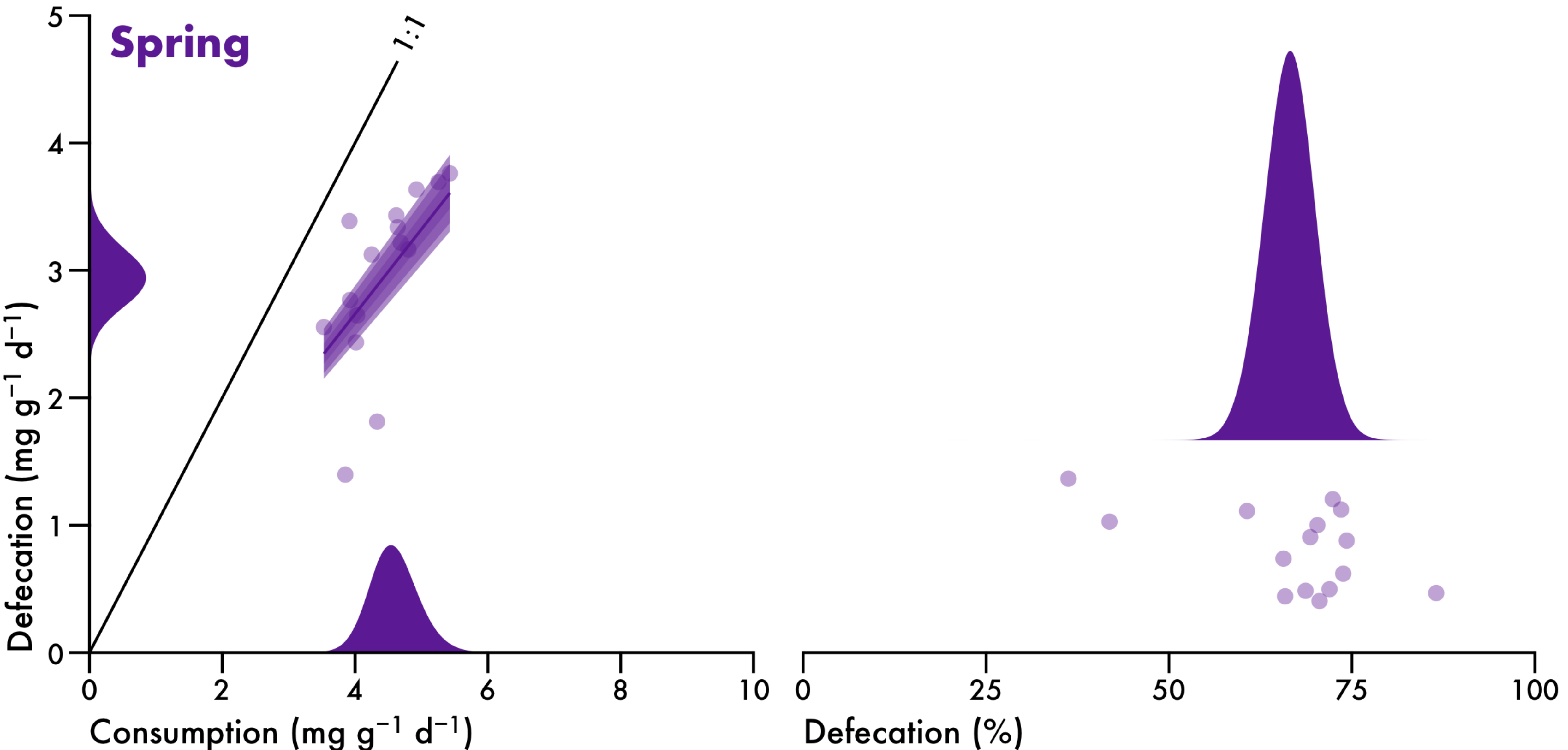
1. Biomass



1. Biomass

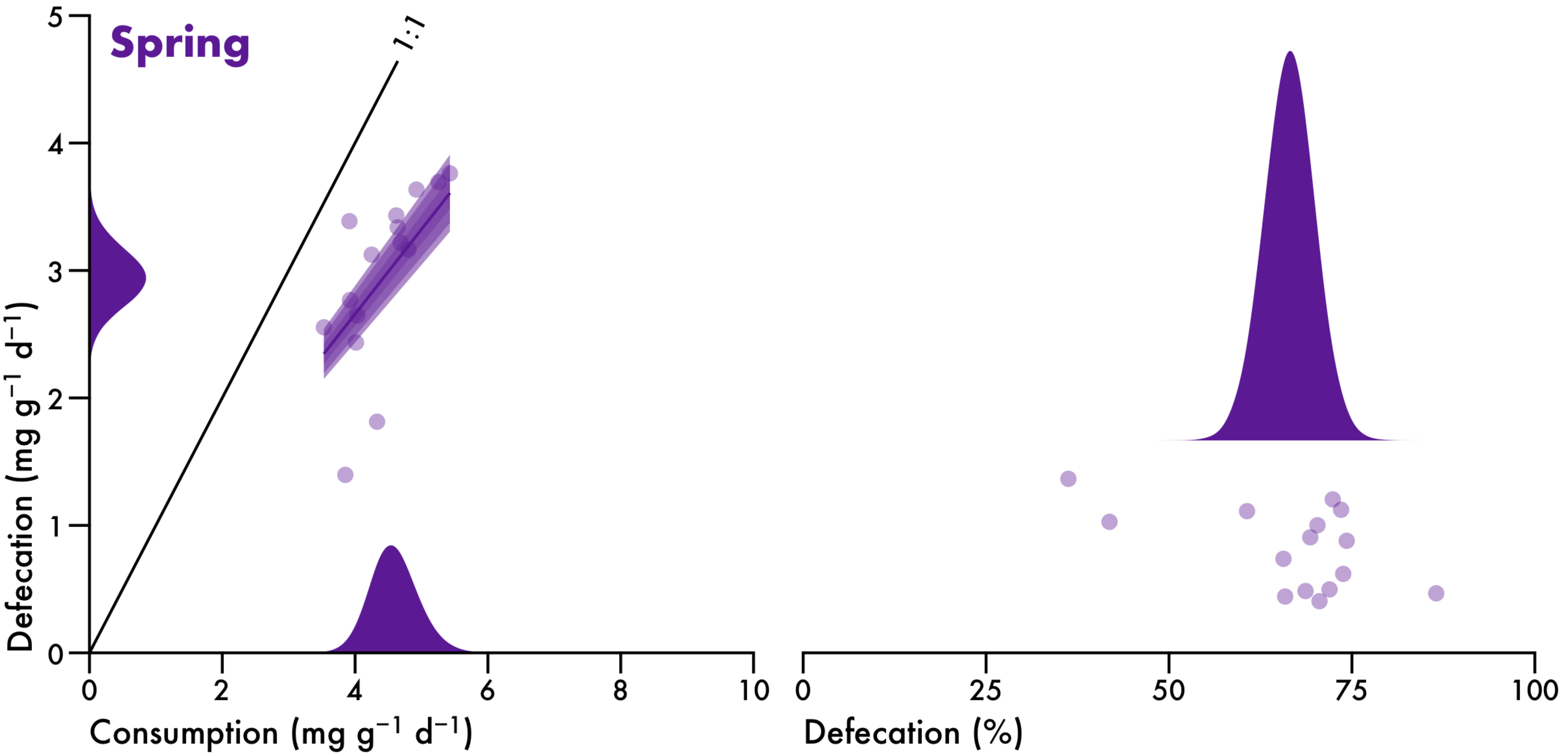


1. Biomass



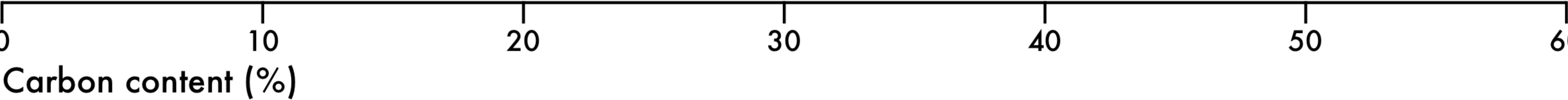
1. Biomass

Annual: $44 \pm 20\%$



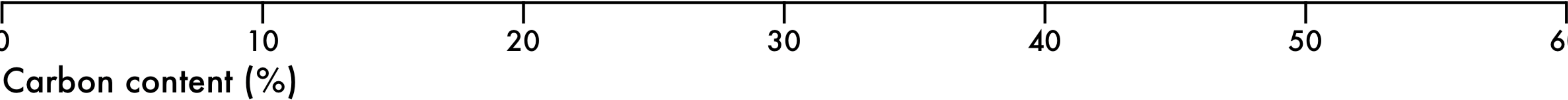
2. Carbon

Prior Kelp Faeces



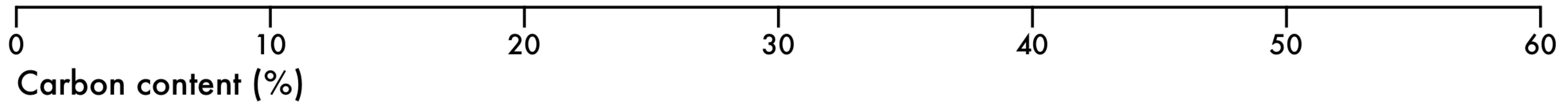
2. Carbon

Prior Kelp Faeces



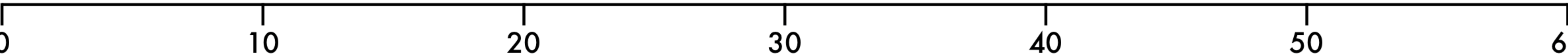
2. Carbon

■ Prior ■ Kelp ■ Faeces

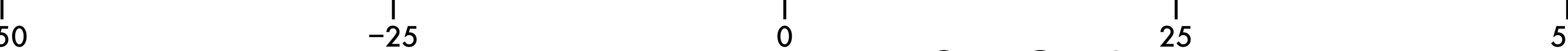
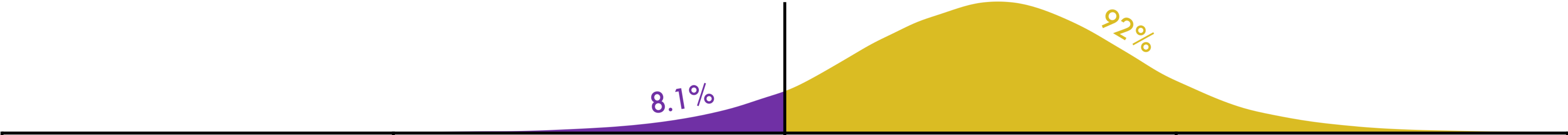


2. Carbon

Prior Kelp Faeces



Carbon content (%)



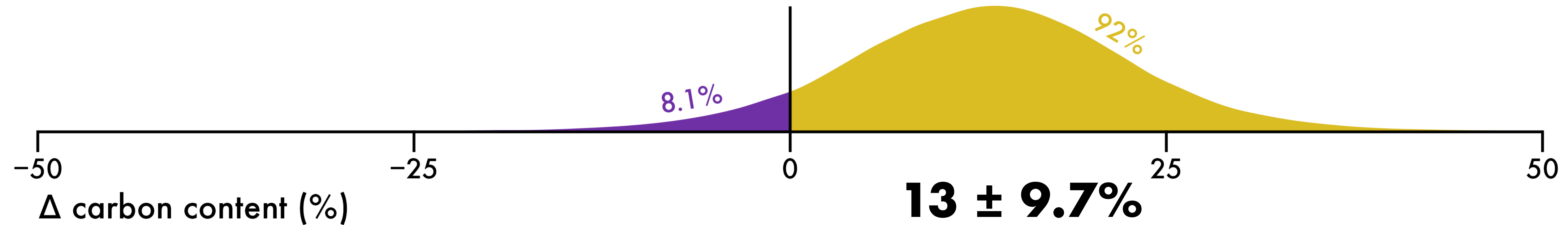
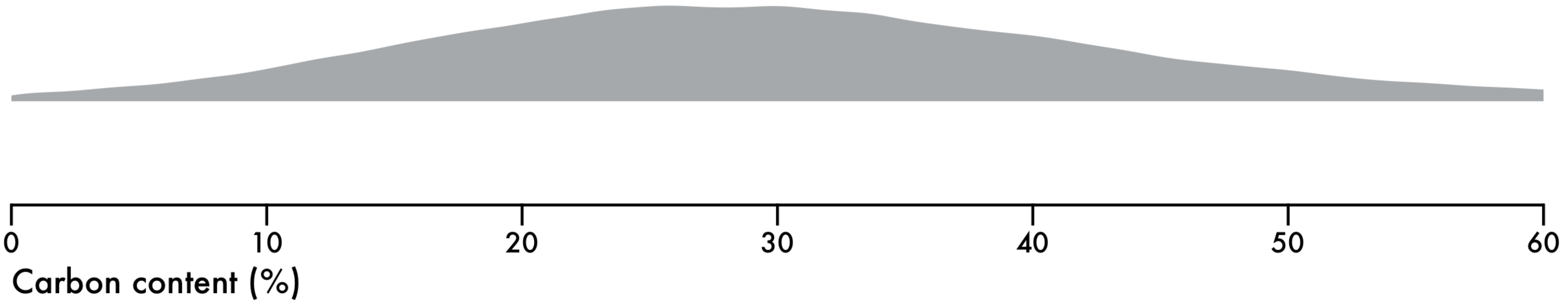
Δ carbon content (%)

13 ± 9.7%

2. Carbon

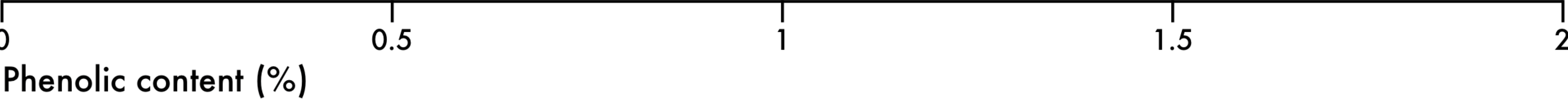
■ Prior ■ Kelp ■ Faeces

$$\text{Faeces} = 0.65 \pm 0.27 \times \text{Kelp}$$



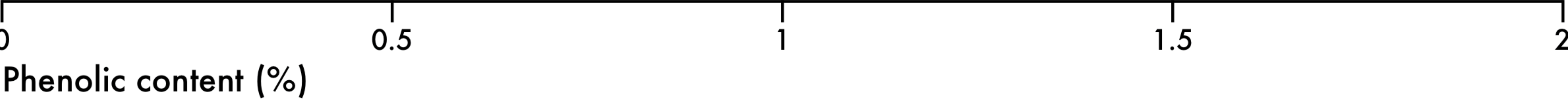
3. Chemical defence

■ Prior ■ Kelp ■ Faeces



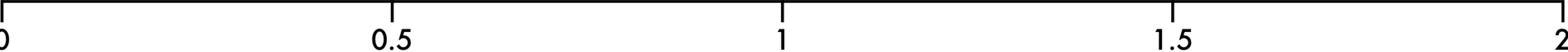
3. Chemical defence

■ Prior ■ Kelp ■ Faeces

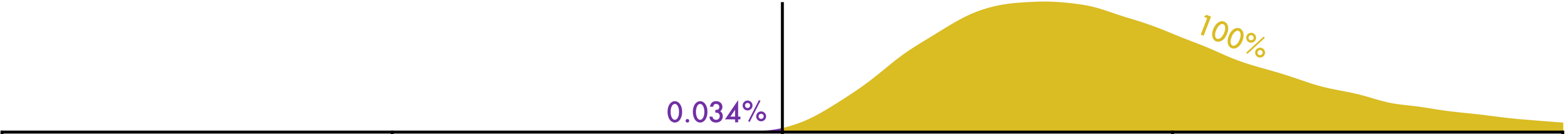


3. Chemical defence

Prior Kelp Faeces



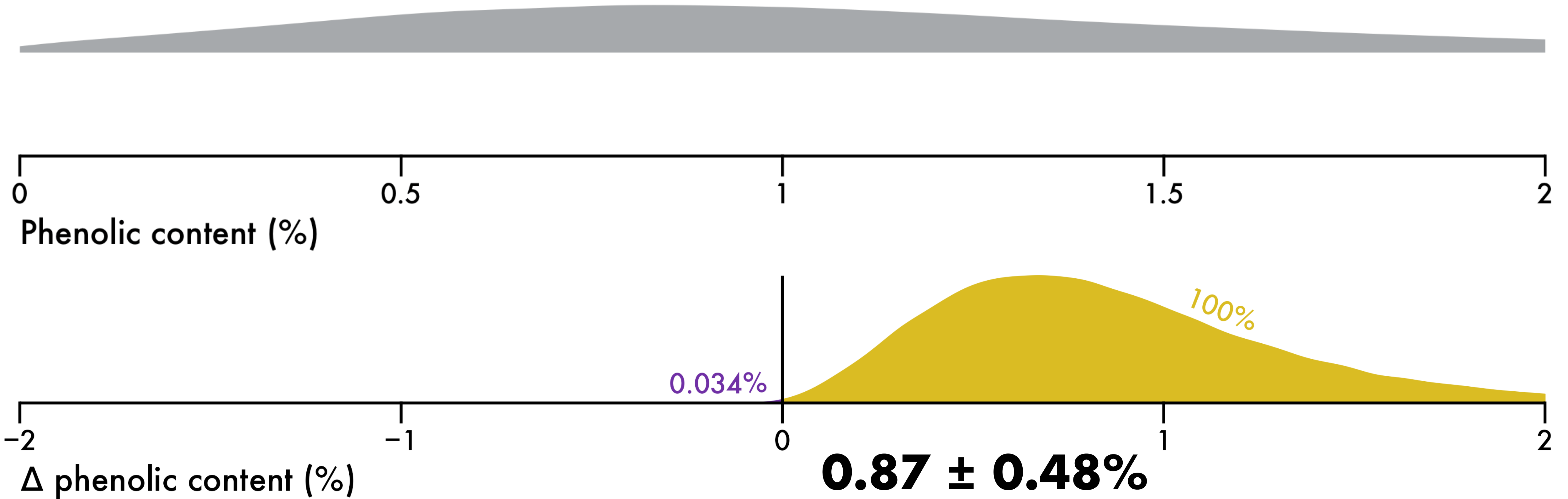
Phenolic content (%)



Δ phenolic content (%)

3. Chemical defence

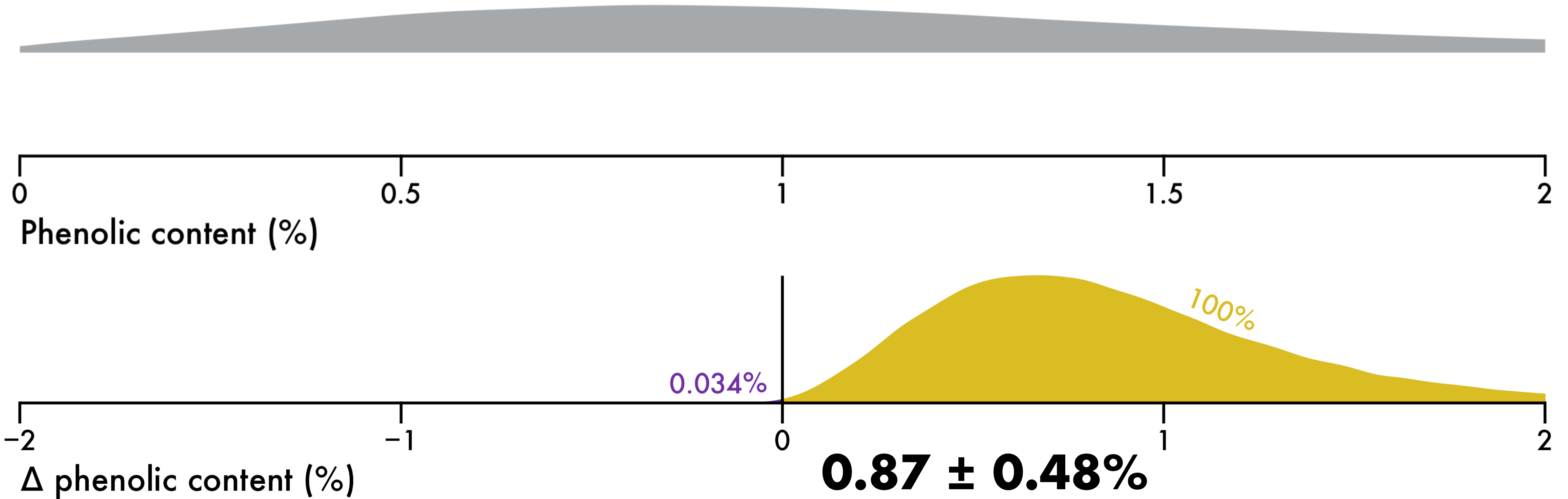
Prior Kelp Faeces



3. Chemical defence

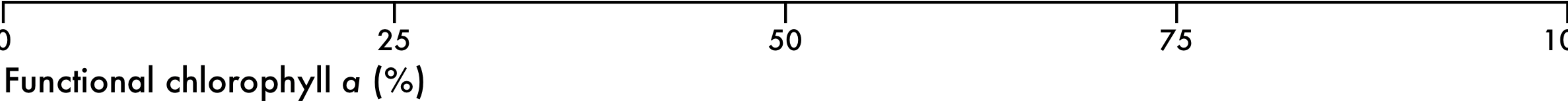
Increased palatability

Prior Kelp Faeces



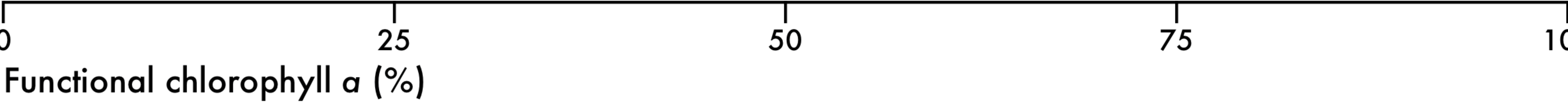
4. Photophysiology

Prior Kelp Faeces



4. Photophysiology

Prior Kelp Faeces



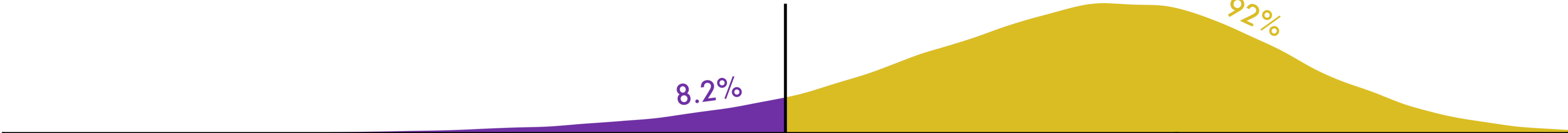
4. Photophysiology

Prior Kelp Faeces



0 25 50 75 100

Functional chlorophyll a (%)



-80 -40 0 40 80

Δ functional chlorophyll a (%)

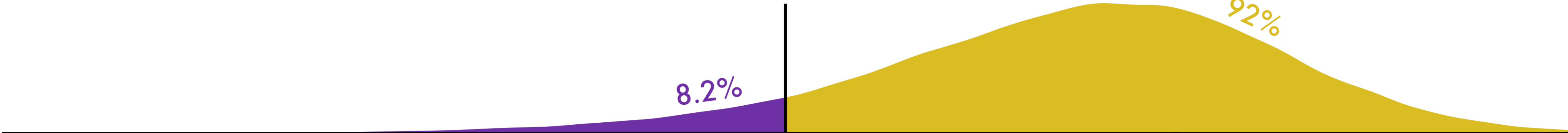
4. Photophysiology

Prior Kelp Faeces



0 25 50 75 100

Functional chlorophyll a (%)



-80 -40 0 40 80

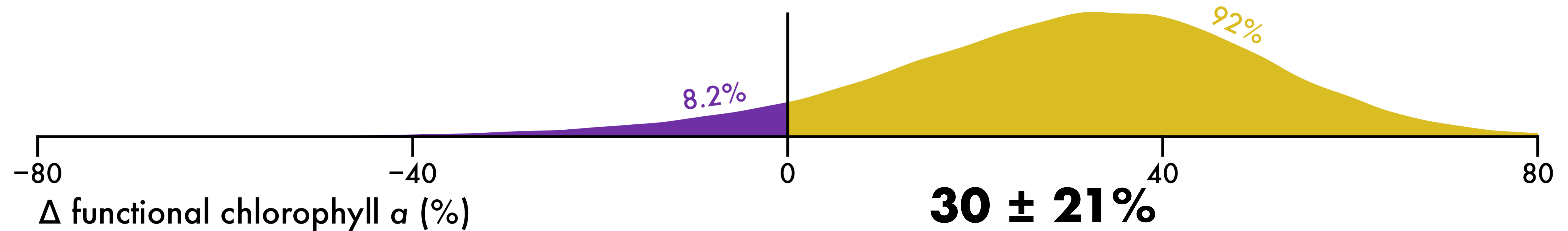
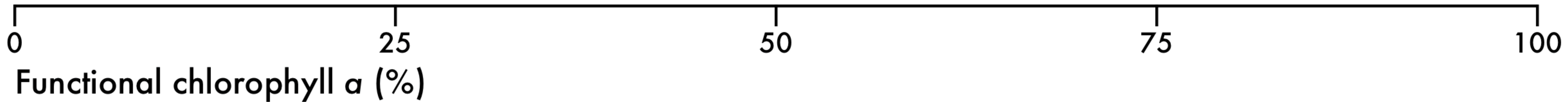
Δ functional chlorophyll a (%)

30 ± 21%

4. Photophysiology

■ Prior ■ Kelp ■ Faeces

No detrital photosynthesis



Findings

Findings

1. Carbon transfer is $\overbrace{0.44 \pm 0.2}^{\text{Defecation}} \times \overbrace{0.65 \pm 0.27}^{\text{Carbon}} = 0.29 \pm 0.19$

Findings

Defecation

Carbon

1. Carbon transfer is $0.44 \pm 0.2 \times 0.65 \pm 0.27 = 0.29 \pm 0.19$
2. Urchin faeces likely decompose much faster

Answer

Do sea urchins really extend kelp carbon transfer?

Answer

**Do sea urchins really extend kelp carbon transfer?
No. Probably the opposite.**

Answer

Do sea urchins really extend kelp carbon transfer?

No. Probably the opposite.

Relative carbon sequestration potential of urchin faeces is
 $0.29 \pm 0.19 \times 2.3 \pm 0.14 = 0.67 \pm 0.44$ (no decomposition).

Transfer

Export

Contact

 0000-0002-1273-6256

 @lukaseamus

 @LukaSeaWright

 Luka Seamus Wright

 luka@wright.it

